WORKED SOLUTION

STEP III 2019, Question 4

Pure •

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Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

The alternating signs are exactly the signs in Vieta's formulas: a monic polynomial with roots $r_1, \ldots, r_n$ is $x^n - e_1 x^{n-1} + e_2 x^{n-2} - \cdots + (-1)^n e_n$, where e_k is the sum of all products of k of the roots. So "reflexive" just says that $e_k(a_1, \ldots, a_n) = a_k$ for every k. Small cases are then a few simultaneous equations, and the sum of squares of the roots, $e_1^2 - 2e_2$, gives the identity in (ii). Once $a_n = 0$ is forced, a factor of x comes out and we can peel the polynomial down.

Solution

Throughout, write e_k for the kth elementary symmetric function of $a_1, \ldots, a_n$. By Vieta, P has roots $a_1, \ldots, a_n$ exactly when

$$e_k(a_1, \ldots, a_n) = a_k \qquad \text{for } k = 1, 2, \ldots, n.$$

Part (i)

Degree 1. $\mathrm{P}(x) = x - a_1$ has root a_1, so every such polynomial is reflexive: $\mathrm{P}(x) = x - a$ for any real a.

Degree 2. We need $a_1 + a_2 = a_1$ and $a_1 a_2 = a_2$. The first gives $a_2 = 0$, and then the second holds automatically. So $\mathrm{P}(x) = x^2 - ax = x(x - a)$ for any real a; its roots a and 0 are indeed a_1 and a_2.

Degree 3. We need

$$\begin{aligned} a_1 + a_2 + a_3 &= a_1, \\ a_1 a_2 + a_1 a_3 + a_2 a_3 &= a_2, \\ a_1 a_2 a_3 &= a_3. \end{aligned}$$

The first gives $a_3 = -a_2$. Then the second becomes $a_1(a_2 + a_3) + a_2 a_3 = a_2$, that is $-a_2^2 = a_2$, so $a_2 = 0$ or $a_2 = -1$.

- If $a_2 = 0$ then $a_3 = 0$ and the third equation reads $0 = 0$. So a_1 is arbitrary: $\mathrm{P}(x) = x^3 - ax^2 = x^2(x-a)$.
- If $a_2 = -1$ then $a_3 = 1$ and the third equation gives $-a_1 = 1$, so $a_1 = -1$. This gives

$$\mathrm{P}(x) = x^3 + x^2 - x - 1 = (x+1)^2(x-1),$$

whose roots $-1, -1, 1$ are indeed a_1, a_2, a_3.

So the reflexive polynomials of degree at most 3 are

$$x - a, \qquad x^2 - ax, \qquad x^3 - ax^2 \quad (a \in \mathbb{R}), \qquad \text{and} \qquad x^3 + x^2 - x - 1.$$

Part (ii)

The sum of the squares of the roots is

$$a_1^2 + a_2^2 + \cdots + a_n^2 = e_1^2 - 2e_2,$$

since squaring e_1 gives each a_i^2 once and each product $a_i a_j$ $(i < j)$ twice. For a reflexive polynomial $e_1 = a_1$ and $e_2 = a_2$, so

$$a_1^2 + a_2^2 + \cdots + a_n^2 = a_1^2 - 2a_2, \qquad \text{that is} \qquad 2a_2 = -a_2^2 - a_3^2 - \cdots - a_n^2,$$

as required. (This works for every $n \geq 2$.)

Now suppose the a_k are integers, $a_n \neq 0$ and, for contradiction, $n > 3$. Moving a_2^2 across and completing the square,

$$(a_2 + 1)^2 = 1 - \left(a_3^2 + a_4^2 + \cdots + a_n^2\right).$$

? **Ask yourself** How do integers squeeze this equation?

The left side is at least 0, so $a_3^2 + \cdots + a_n^2 \leq 1$. But a_n is a non-zero integer, so $a_n^2 \geq 1$. Hence the sum is exactly 1: $a_n^2 = 1$ and every other term is zero, so $a_3 = a_4 = \cdots = a_{n-1} = 0$, and also $a_2 = -1$. Since $n \geq 4$, the list $a_3, \ldots, a_{n-1}$ is not empty, so in particular $a_3 = 0$. Now the last Vieta equation says the product of the roots is a_n:

$$a_n = e_n = a_1 a_2 a_3 \cdots a_n = 0,$$

because $a_3 = 0$. This contradicts $a_n \neq 0$. Therefore $n \leq 3$.

Part (iii)

Removing a factor of x. Suppose P is reflexive of degree $n \geq 2$ with $a_n = 0$. Then $\mathrm{P}(x) = x\,\mathrm{Q}(x)$ with

$$\mathrm{Q}(x) = x^{n-1} - a_1 x^{n-2} + \cdots + (-1)^{n-1} a_{n-1}.$$

The roots of P are $a_1, \ldots, a_{n-1}, a_n$ with $a_n = 0$, and the factor x accounts for one root 0. So the roots of Q are the rest, $a_1, \ldots, a_{n-1}$: Q is reflexive, with the same (integer)

coefficients. Conversely, if Q is reflexive of degree m with coefficients $a_1, \ldots, a_m$, then $x\,\mathrm{Q}(x)$ has coefficients $a_1, \ldots, a_m, 0$ and roots $a_1, \ldots, a_m, 0$, so it is reflexive too.

Peeling down. Start with any reflexive P with integer coefficients. While its degree is at least 2 and its last coefficient is 0, remove a factor of x. This stops at $\mathrm{P}(x) = x^m \mathrm{R}(x)$, with $m \geq 0$ and R an integer reflexive polynomial of degree k that either has $k = 1$ or has non-zero last coefficient a_k.

- If $a_k \neq 0$, then $k \leq 3$ by part (ii) (for $k \leq 3$ there is nothing to prove). From the list in part (i): with $k = 1$ we get $x - a$ with $a \neq 0$; with $k = 2$ the last coefficient is always 0, so nothing; with $k = 3$ the only one with $a_3 \neq 0$ is $x^3 + x^2 - x - 1$.
- If $k = 1$ and $a_1 = 0$, then $\mathrm{R}(x) = x$.

Folding $\mathrm{R}(x) = x$ into the case $x - a$ with $a = 0$, and using the converse above to see that every multiple by x^m is reflexive, the reflexive polynomials with integer coefficients are exactly

$$x^m(x - a) \quad (m \geq 0,\ a \in \mathbb{Z}),$$
$$x^m(x^3 + x^2 - x - 1) = x^m(x + 1)^2(x - 1) \quad (m \geq 0).$$

Checks

- $x^2(x+1)^2(x-1) = x^5 + x^4 - x^3 - x^2$ has $a_1, \ldots, a_5 = -1, -1, 1, 0, 0$ and roots $-1, -1, 1, 0, 0$: it is reflexive. The identity in (ii) reads $2(-1) = -(1 + 1 + 0 + 0)$, which holds.
- A computer search over every integer choice of $a_1, \ldots, a_n$ in $\{-3, \ldots, 3\}$ with $n \leq 7$ found no reflexive polynomials other than those listed; a search over small rationals for $n = 2, 3$ confirmed the list in part (i).

Common slips

- Forgetting the families with a free parameter: $x^2 - ax$ and $x^3 - ax^2$ are reflexive for every real a, not just for special values.
- In (ii), finding $a_n = \pm 1$, $a_2 = -1$ and stopping. The contradiction comes from the product of the roots, which vanishes because some $a_3, \ldots, a_{n-1}$ is zero when $n \geq 4$.
- In (iii), saying "$a_n = 0$ so divide by x" without checking that the quotient is again reflexive: one zero root goes with the factor x, and the remaining roots match the remaining coefficients.

Beyond STEP

The equations $e_k(a_1, \ldots, a_n) = a_k$ are a small example of a polynomial system whose solutions form an algebraic variety, here a union of lines and isolated points. The fact

that "sum of squares $= e_1^2 - 2e_2$" is the first of Newton's identities, which link power sums of the roots to the coefficients and underlie the theory of symmetric functions. The step "integers that are squares and sum to at most 1" is a typical move in Diophantine arguments. Davenport's *The Higher Arithmetic* and Allenby's *Rings, Fields and Groups* are good places to explore these ideas further.

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